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MTH501 - Linear Algebra (Quiz 3) Quiz Start Time: 01:45 PM, 24 July 2023

Question # 1 of 10 (Start time: 01:45:43 PM, 24 July 2023) Total Marks: 1

What is the maximum possible number of pivots in a 4x6 matrix ?

Select the correct option

Reload Math Equations

- 4
- 8
- 10
- 6

Click to Save Answer & Move to Next Question

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MTH501 - Linear Algebra (Quiz 3)

Quiz Start Time: 01:45 PM, 24 July 2023

Question # 2 of 10 (Start time: 01:46:22 PM, 24 July 2023)

Total Marks: 1

Let 'Ax = 0' be a homogeneous linear system of 'n' equations and 'n' unknowns. Then, the coefficient matrix 'A' is invertible if and only if this system has _____ solution.

Select the correct option

Reload Math Equations

- non-trivial
- infinite many
- no
- trivial

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MTH501 - Linear Algebra (Quiz 3) Quiz Start Time: 01:45 PM, 24 July 2023

Question # 3 of 10 (Start time: 01:47:51 PM, 24 July 2023) Total Marks: 1

For u, v vectors in R^n, the distance between u and v is written as

Select the correct option

Reload Math Equations

- None of these
- $dist(u, v) = \|u - v\|$
- $dist(u, v) = \|u - u\|$
- $dist(u, v) = \|v - v\|$

Click to Save Answer & Move to Next Question

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MTH501 - Linear Algebra (Quiz 3)

Quiz Start Time: 01:45 PM, 24 July 2023

Question # 4 of 10 (Start time: 01:48:47 PM, 24 July 2023)

Total Marks: 1

Elementary row operations on a matrix do not affect the --------- relations among the columns of the matrix.

Select the correct option

Reload Math Equations

- Linear dependence
- Linear independence

Click to Save Answer & Move to Next Question

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MTH501 - Linear Algebra (Quiz 3)

Quiz Start Time: 01:45 PM, 24 July 2023

Question # 5 of 10 (Start time: 01:49:51 PM, 24 July 2023)

Total Marks: 1

An n x n matrix A is _____ if and only if A has n linearly independent vectors.

Select the correct option

- scalar
- diagonalizable
- singular
- symmetric

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MTH501 - Linear Algebra (Quiz 3) Quiz Start Time: 01:45 PM, 24 July 2023

Question # 6 of 10 (Start time: 01:50:48 PM, 24 July 2023) Total Marks: 1

We can add the matrices of _____

Select the correct option

Reload Math Equations

- same number of columns
- same order
- same number of rows
- different order

Click to Save Answer & Move to Next Question

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Question # 7 of 10 (Start time: 01:51:35 PM, 24 July 2023) Total Marks: 1

Which of the following is true for the matrix

$$M = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ O & M_{22} & M_{23} \\ O & O & M_{33} \end{bmatrix};$$

where M_{11}, M_{22} and M_{33} are square sub-matrices, and O is a zero sub-matrix ?

Select the correct option

Reload Math Equations

- It is a block lower triangular matrix.
- It is a Null matrix.
- It is a block upper triangular matrix.
- It is diagonal-constant matrix.

Click to Save Answer & Move to Next Question

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MTH501 - Linear Algebra (Quiz 3) Quiz Start Time: 01:45 PM, 24 July 2023

Question # 8 of 10 (Start time: 01:52:47 PM, 24 July 2023) Total Marks: 1

What is Eigen value?

Select the correct option

- A scalar associated with a given linear transformation
- A matrix determined from the algebraic equations
- A vector obtained from the coordinates
- It is the inverse of the transform

Question # 10 of 10 (Start time: 01:54:46 PM, 24 July 2023)

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Total Marks: 1

When a matrix A has a complex eigenvalue with positive real part, the trajectories spiral ----------

Select the correct option

Reload Math Equations

- Inwards
- Outwards

Click to Save Answer & Move to Next Question

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Question # 9 of 10 (Start time: 01:53:55 PM, 24 July 2023) Total Marks: 1

An n x n matrix A is invertible if and only if _____ is not an eigen value of A.

Select the correct option

- 2
- 0
- -1
- 1

Click to Save Answer & Move to Next Question